

# Inventories and Cost of Sales

## Chapter 6

# Chapter 6 Learning Objectives

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## CONCEPTUAL

- C1** Identify the items making up merchandise inventory.
- C2** Identify the costs of merchandise inventory.

## ANALYTICAL

- A1** Analyze the effects of inventory methods for both financial and tax reporting.
- A2** Analyze the effects of inventory errors on current and future financial statements.
- A3** Assess inventory management using both inventory turnover and days' sales in inventory.

## PROCEDURAL

- P1** Compute inventory in a perpetual system using the methods of specific identification, FIFO, LIFO, and weighted average.
- P2** Compute the lower of cost or market amount of inventory.
- P3** [Appendix 6A](#)—Compute inventory in a periodic system using the methods of specific identification, FIFO, LIFO, and weighted average.
- P4** [Appendix 6B](#)—Apply both the retail inventory and gross profit methods to estimate inventory.

# Learning Objective

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C1:

Identify the items making up  
merchandise inventory.

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# Determining Inventory Items

Merchandise inventory includes all goods that a company owns and holds for sale, regardless of where the goods are located when inventory is counted.

Items requiring special attention include:

Goods in  
Transit

Goods on  
Consignment

Goods  
Damaged or  
Obsolete

# Goods in Transit



# Goods on Consignment

- Merchandise is included in the inventory of the consignor, the owner of the inventory.
- Consignee sells goods for the owner, consignor.
- Consignee never reports consigned goods in inventory.

# Goods Damaged or Obsolete

Damaged or obsolete goods are not reported in inventory if they cannot be sold.

Cost should be reduced to net realizable value if they can be sold.

# Learning Objective

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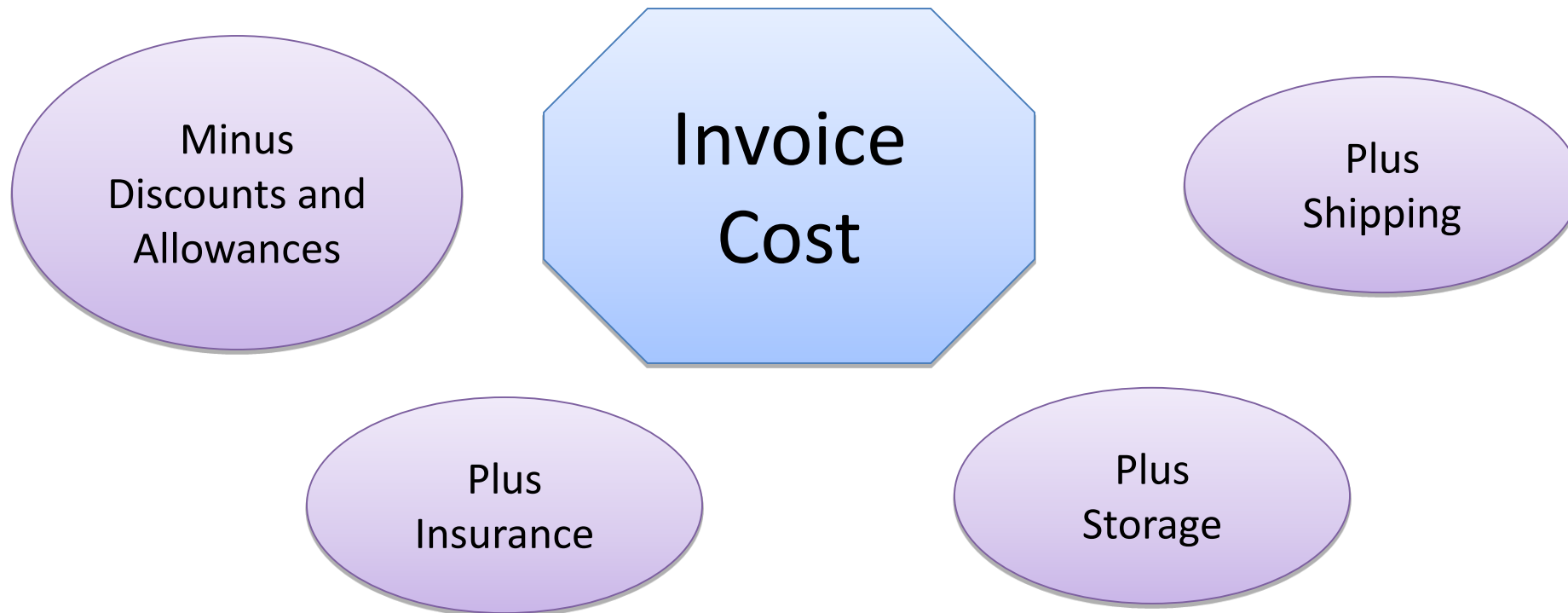
C2:

Identify the costs of  
merchandise inventory.

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# Determining Inventory Costs

Include all expenditures necessary to bring an item to a salable condition and location.



# Internal Controls and Taking a Physical Count

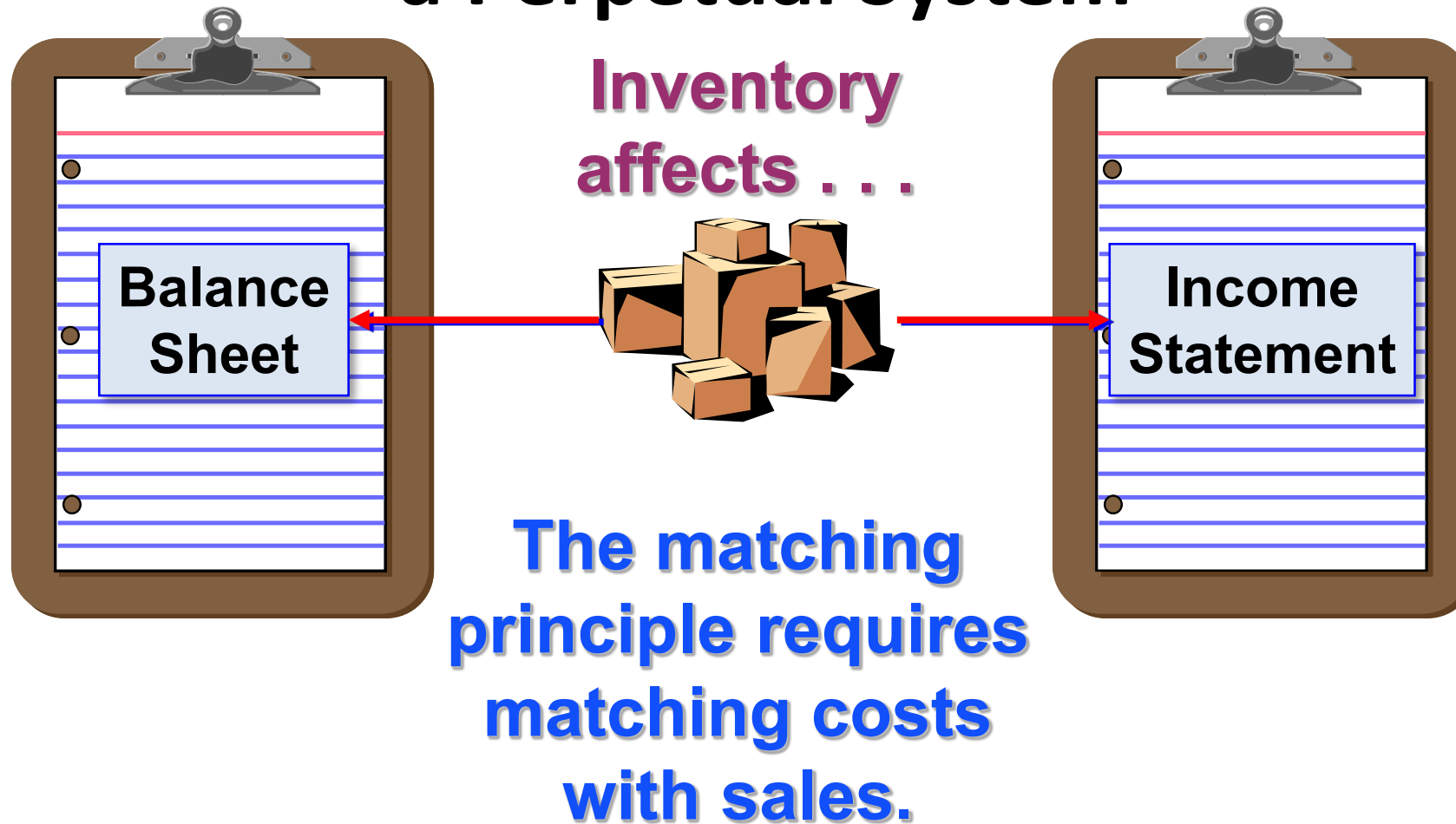
➤ Most companies take a physical count of inventory at least once each year.

➤ When the physical count does not match the Merchandise Inventory account, an adjustment must be made.

## **Good internal controls over count include:**

1. Pre-numbered inventory tickets.
2. Counters have no inventory responsibility.
3. Counters confirm existence, amount, and quality of inventory item.
4. Second count is taken.
5. Manager confirms all items counted.

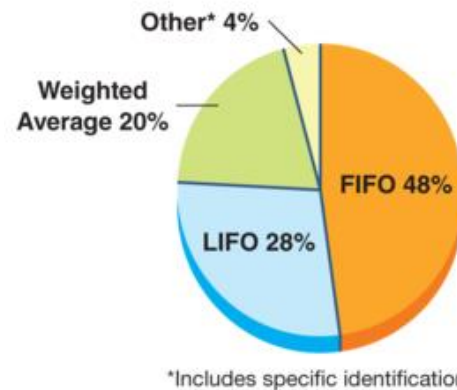
# Inventory Costing under a Perpetual System



# Inventory Costing Management Decisions

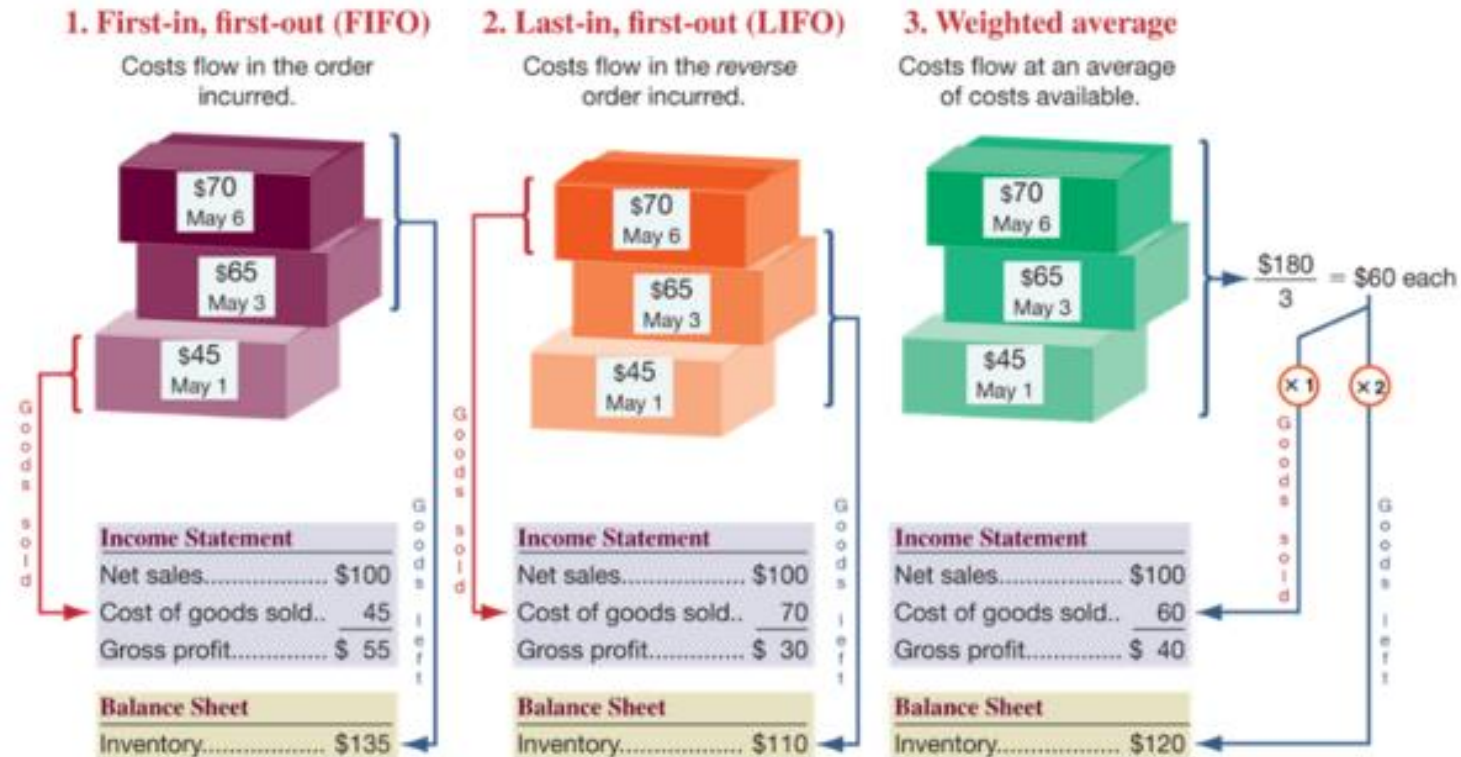
Management decisions in accounting for inventory involve the following:

1. Items included in inventory and their costs.
2. Costing method (specific identification, FIFO, LIFO, or weighted average).
3. Inventory system (perpetual or periodic).
4. Use of market values or other estimates.

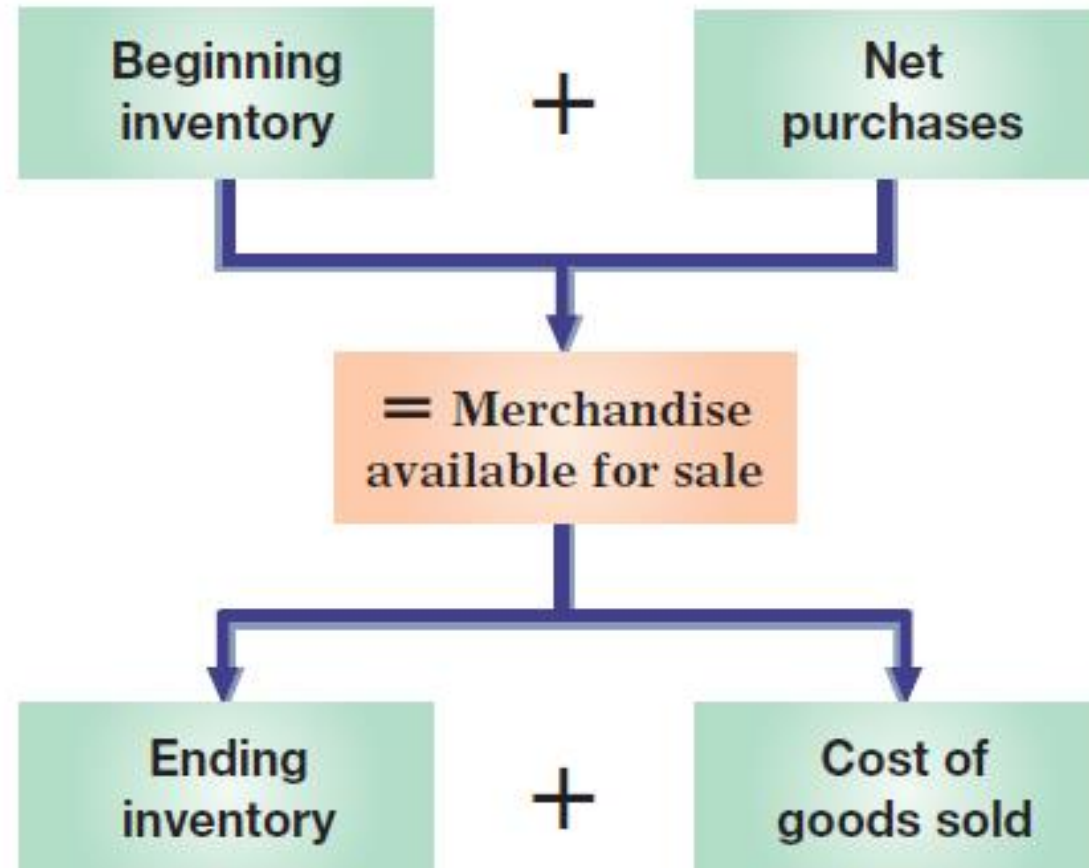


# Inventory Cost Flow Assumptions

Exhibit  
6.2



# Inventory Cost Flow Assumptions



# Learning Objective

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P1:

Compute inventory in a perpetual system using the methods of specific identification, FIFO, LIFO and Weighted Average.

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# Inventory Costing Illustration

Exhibit  
6.3

Here is information about the mountain bike inventory of Trekking for the month of August.

| Date    | Activity                 | Units Acquired at Cost                       | Units Sold at Retail     | Unit Inventory             |
|---------|--------------------------|----------------------------------------------|--------------------------|----------------------------|
| Aug. 1  | Beginning inventory..... | 10 units @ \$ 91 = \$ 910                    |                          | 10 units                   |
| Aug. 3  | Purchases.....           | 15 units @ \$106 = \$ 1,590                  |                          | 25 units                   |
| Aug. 14 | Sales.....               |                                              | 20 units @ \$130         | 5 units                    |
| Aug. 17 | Purchases.....           | 20 units @ \$115 = \$ 2,300                  |                          | 25 units                   |
| Aug. 28 | Purchases.....           | 10 units @ \$119 = \$ 1,190                  |                          | 35 units                   |
| Aug. 30 | Sales.....               |                                              | 23 units @ \$150         | <b>12 units</b>            |
|         | Totals .....             | <u><b>55 units</b></u> <u><b>\$5,990</b></u> | <u><b>43 units</b></u>   |                            |
|         |                          | Units available for sale                     | Goods available for sale | Units sold      Units left |

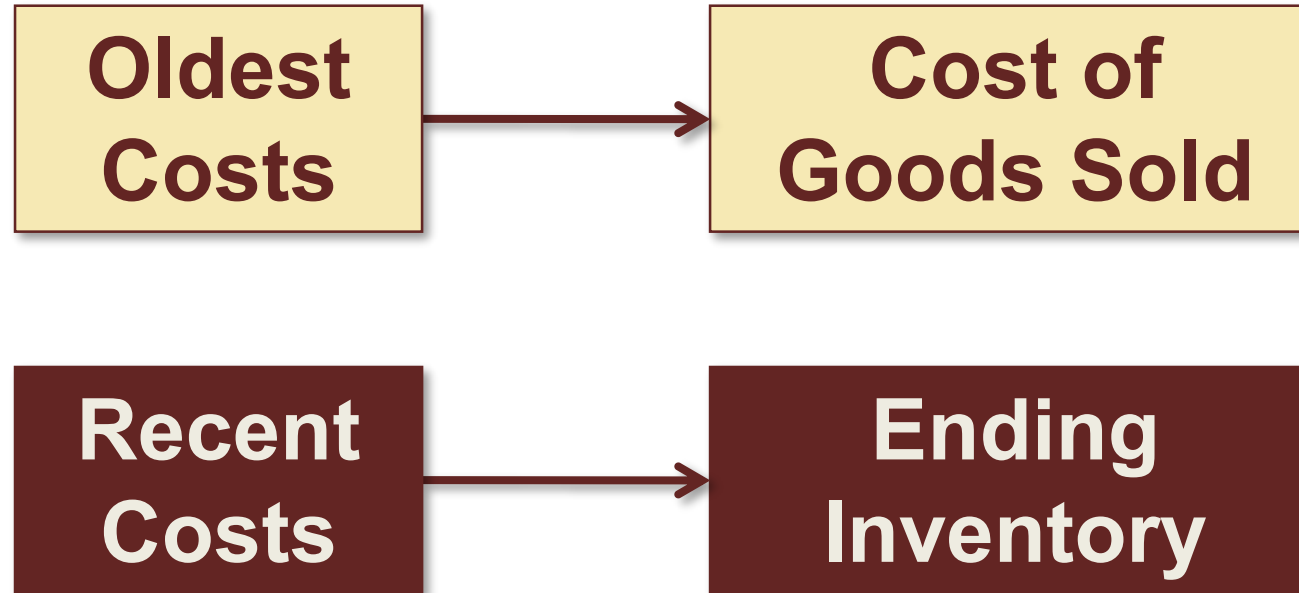
# Specific Identification

Exhibit  
6.4

|                                                                     |              |                       |
|---------------------------------------------------------------------|--------------|-----------------------|
| Total cost of 55 units available for sale (from Exhibit 6.3) .....  |              | \$ 5,990              |
| <b>Cost of goods sold*</b>                                          |              |                       |
| Aug. 14 (8 @ \$91) + (12 @ \$106).....                              | \$2,000      |                       |
| Aug. 30 (2 @ \$91) + (3 @ \$106) + (15 @ \$115) + (3 @ \$119) ..... | <u>2,582</u> | <u>4,582</u>          |
| <b>Ending inventory</b> .....                                       |              | <u><u>\$1,408</u></u> |

\* Identification of items sold (and their costs) is from internal documents that track each unit from its purchase to its sale.

# First-In, First-Out (FIFO)



# First-In, First-Out (FIFO)

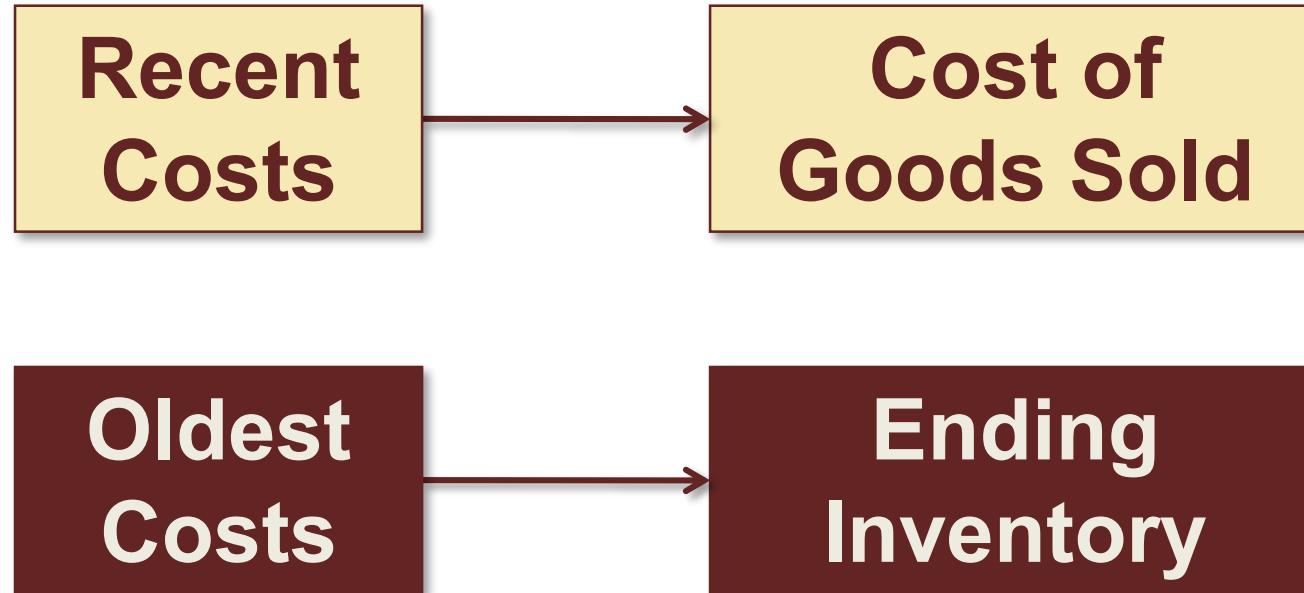
Exhibit  
6.5

For the 20 units sold on Aug. 14, the first 10 sold are assigned the earliest cost of \$91 (from beg. bal.). The next 10 sold are assigned the next earliest cost of \$106.

For the 23 units sold on Aug. 30, the first 5 sold are assigned the earliest available cost of \$106 (from Aug. 3 purchase). The next 18 sold are assigned the next earliest cost of \$115 (from Aug. 17 purchase).

| Date    | Goods Purchased      | Cost of Goods Sold                                               | Inventory Balance                                      |
|---------|----------------------|------------------------------------------------------------------|--------------------------------------------------------|
| Aug. 1  | Beginning balance    |                                                                  | 10 @ \$ 91 = \$ 910                                    |
| Aug. 3  | 15 @ \$106 = \$1,590 |                                                                  | 10 @ \$ 91 }<br>15 @ \$106 } = \$ 2,500                |
| Aug. 14 |                      | 10 @ \$ 91 = \$ 910 }<br>10 @ \$106 = \$1,060 } = <b>\$1,970</b> | 5 @ \$106 = \$ 530                                     |
| Aug. 17 | 20 @ \$115 = \$2,300 |                                                                  | 5 @ \$106 }<br>20 @ \$115 } = \$ 2,830                 |
| Aug. 28 | 10 @ \$119 = \$1,190 |                                                                  | 5 @ \$106 }<br>20 @ \$115 }<br>10 @ \$119 } = \$ 4,020 |
| Aug. 30 |                      | 5 @ \$106 = \$ 530 }<br>18 @ \$115 = \$2,070 } = <b>\$2,600</b>  | 2 @ \$115 }<br>10 @ \$119 } = <b><u>\$1,420</u></b>    |
|         |                      | <b><u>\$4,570</u></b>                                            |                                                        |

# Last-In, First-Out (LIFO)



# Last-In, First-Out (LIFO)

Exhibit  
6.6

| Date    | Goods Purchased      | Cost of Goods Sold                                                                                                                  | Inventory Balance                                                                          |
|---------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Aug. 1  | Beginning balance    |                                                                                                                                     | 10 @ \$ 91 = \$ 910                                                                        |
| Aug. 3  | 15 @ \$106 = \$1,590 |                                                                                                                                     | <div> <div>10 @ \$ 91</div> <div>15 @ \$106</div> </div> } = \$ 2,500                      |
| Aug. 14 |                      | <div> <div>15 @ \$106 = \$1,590</div> <div>5 @ \$ 91 = \$ 455</div> </div> } = <b>\$2,045</b>                                       | <div> <div>5 @ \$ 91</div> </div> } = \$ 455                                               |
| Aug. 17 | 20 @ \$115 = \$2,300 |                                                                                                                                     | <div> <div>5 @ \$ 91</div> <div>20 @ \$115</div> </div> } = \$ 2,755                       |
| Aug. 28 | 10 @ \$119 = \$1,190 |                                                                                                                                     | <div> <div>5 @ \$ 91</div> <div>20 @ \$115</div> <div>10 @ \$119</div> </div> } = \$ 3,945 |
| Aug. 30 |                      | <div> <div>10 @ \$119 = \$1,190</div> <div>13 @ \$115 = \$1,495</div> </div> } = <b>\$2,685</b><br><div><b><u>\$4,730</u></b></div> | <div> <div>5 @ \$ 91</div> <div>7 @ \$115</div> </div> } = <b><u>\$1,260</u></b>           |

For the 20 units sold on Aug. 14, the first 15 sold are assigned the most recent cost of \$106. The next 5 sold are assigned the next most recent cost of \$91.

For the 23 units sold on Aug. 30, the first 10 sold are assigned the most recent cost of \$119. The next 13 sold are assigned the next most recent cost of \$115.

# Weighted Average

When a unit is sold, the **average cost** of each unit in inventory is assigned to cost of goods sold.

$$\frac{\text{Cost of Goods Available for Sale}}{\text{Units on hand on the date of sale}}$$

# Weighted Average

| Date    | Goods Purchased      | Cost of Goods Sold          | Inventory Balance                                                                                     |
|---------|----------------------|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Aug. 1  | Beginning balance    |                             | 10 @ \$ 91 = \$ 910                                                                                   |
| Aug. 3  | 15 @ \$106 = \$1,590 |                             | <div> <div>10 @ \$ 91</div> <div>15 @ \$106</div> </div> } = \$2,500 (or \$100 per unit) <sup>a</sup> |
| Aug. 14 |                      | 20 @ \$100 = <b>\$2,000</b> | 5 @ \$100 = \$ 500 (or \$100 per unit) <sup>b</sup>                                                   |

On August 14, 20 bikes are sold. To determine the cost of the units sold, we first, need to compute the weighted average cost per unit of items in inventory.

|                                  |               |
|----------------------------------|---------------|
| Cost of goods available for sale | \$ 2,500      |
| Units available at time of sale  | ÷ 25          |
| Weighted average cost per unit   | <u>\$ 100</u> |

The cost of goods sold for the August 14 sale is **\$2,000**. After this sale, there are five \$100 units in inventory totaling \$500.

# Weighted Average

| Date    | Goods Purchased      | Cost of Goods Sold          | Inventory Balance                                                                                     |
|---------|----------------------|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Aug. 1  | Beginning balance    |                             | 10 @ \$ 91 = \$ 910                                                                                   |
| Aug. 3  | 15 @ \$106 = \$1,590 |                             | <div> <div>10 @ \$ 91</div> <div>15 @ \$106</div> </div> } = \$2,500 (or \$100 per unit) <sup>a</sup> |
| Aug. 14 |                      | 20 @ \$100 = <b>\$2,000</b> | 5 @ \$100 = \$ 500 (or \$100 per unit) <sup>b</sup>                                                   |
| Aug. 17 | 20 @ \$115 = \$2,300 |                             | <div> <div>5 @ \$100</div> <div>20 @ \$115</div> </div> } = \$2,800 (or \$112 per unit) <sup>c</sup>  |

## August 17:

|                                               |                      |
|-----------------------------------------------|----------------------|
| <b>Cost of goods available for sale</b>       | <b>\$ 2,800</b>      |
| <b>Units available at time of sale 5 + 20</b> | <b>÷ 25</b>          |
| <b>Weighted average cost per unit</b>         | <b><u>\$ 112</u></b> |

After the August 14 sale, there are 5 units in inventory totaling \$500. On August 17, 20 units are purchased for \$2,300.  $\$2,300 + \$500 = \$2,800 / 25 = \$112$  per unit.

# Weighted Average

| Date    | Goods Purchased      | Cost of Goods Sold          | Inventory Balance                                                         |
|---------|----------------------|-----------------------------|---------------------------------------------------------------------------|
| Aug. 1  | Beginning balance    |                             | 10 @ \$ 91 = \$ 910 (10 @ \$ 91 per unit)                                 |
| Aug. 3  | 15 @ \$106 = \$1,590 |                             | 10 @ \$ 91 }<br>15 @ \$106 } = \$2,500 (25 @ \$100 per unit) <sup>a</sup> |
| Aug. 14 |                      | 20 @ \$100 = <b>\$2,000</b> | 5 @ \$100 = \$ 500 ( 5 @ \$100 per unit) <sup>b</sup>                     |
| Aug. 17 | 20 @ \$115 = \$2,300 |                             | 5 @ \$100 }<br>20 @ \$115 } = \$2,800 (25 @ \$112 per unit) <sup>c</sup>  |
| Aug. 28 | 10 @ \$119 = \$1,190 |                             | 25 @ \$112 }<br>10 @ \$119 } = \$3,990 (35 @ \$114 per unit) <sup>d</sup> |

For the 20 units sold on Aug. 14, the cost assigned is the \$100 average cost per unit from the Inventory Balance column at the time of sale.

## August 28:

|                                                |                 |
|------------------------------------------------|-----------------|
| <b>Cost of goods available for sale</b>        | <b>\$ 3,990</b> |
| <b>Units available at time of sale 5+20+10</b> | <b>35</b>       |
| <b>Weighted average cost per unit</b>          | <b>\$ 114</b>   |

After the August 28 purchase, there are 35 (5+20+10) units in inventory totaling (\$2,800+\$1,190) \$3,990.  $\$3,990 / 35 = \$114$  per unit.

# Weighted Average

Exhibit  
6.7

| Date    | Goods Purchased      | Cost of Goods Sold          | Inventory Balance                                                                                                 |
|---------|----------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Aug. 1  | Beginning balance    |                             | 10 @ \$ 91 = \$ 910 (10 @ \$ 91 per unit)                                                                         |
| Aug. 3  | 15 @ \$106 = \$1,590 |                             | $\left. \begin{array}{l} 10 @ \$ 91 \\ 15 @ \$106 \end{array} \right\} = \$2,500 \text{ (25 @ \$100 per unit)}^a$ |
| Aug. 14 |                      | 20 @ \$100 = <b>\$2,000</b> | 5 @ \$100 = \$ 500 ( 5 @ \$100 per unit) <sup>b</sup>                                                             |
| Aug. 17 | 20 @ \$115 = \$2,300 |                             | $\left. \begin{array}{l} 5 @ \$100 \\ 20 @ \$115 \end{array} \right\} = \$2,800 \text{ (25 @ \$112 per unit)}^c$  |
| Aug. 28 | 10 @ \$119 = \$1,190 |                             | $\left. \begin{array}{l} 25 @ \$112 \\ 10 @ \$119 \end{array} \right\} = \$3,990 \text{ (35 @ \$114 per unit)}^d$ |
| Aug. 30 |                      | 23 @ \$114 = <b>\$2,622</b> | 12 @ \$114 = <b><u>\$1,368</u></b> (12 @ \$114 per unit) <sup>e</sup>                                             |
|         |                      | <b><u>\$4,622</u></b>       |                                                                                                                   |

Cost of goods sold for  
August 30 sale is = **\$2,622**

Ending inventory is  
composed of 12 units @  
an average cost of \$114  
each or **\$1,368.**

# Weighted Average

Exhibit  
6.7

For the 20 units sold on Aug. 14, the cost assigned is the \$100 average cost per unit from the Inventory Balance column at the time of sale.

For the 23 units sold on Aug. 30, the cost assigned is the \$114 average cost per unit from the Inventory Balance column at the time of sale.

| Date    | Goods Purchased      | Cost of Goods Sold          | Inventory Balance                                                                                                 |
|---------|----------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Aug. 1  | Beginning balance    |                             | 10 @ \$ 91 = \$ 910 (10 @ \$ 91 per unit)                                                                         |
| Aug. 3  | 15 @ \$106 = \$1,590 |                             | $\left. \begin{array}{l} 10 @ \$ 91 \\ 15 @ \$106 \end{array} \right\} = \$2,500 \text{ (25 @ \$100 per unit)}^a$ |
| Aug. 14 |                      | 20 @ \$100 = <b>\$2,000</b> | 5 @ \$100 = \$ 500 ( 5 @ \$100 per unit) <sup>b</sup>                                                             |
| Aug. 17 | 20 @ \$115 = \$2,300 |                             | $\left. \begin{array}{l} 5 @ \$100 \\ 20 @ \$115 \end{array} \right\} = \$2,800 \text{ (25 @ \$112 per unit)}^c$  |
| Aug. 28 | 10 @ \$119 = \$1,190 |                             | $\left. \begin{array}{l} 25 @ \$112 \\ 10 @ \$119 \end{array} \right\} = \$3,990 \text{ (35 @ \$114 per unit)}^d$ |
| Aug. 30 |                      | 23 @ \$114 = <b>\$2,622</b> | 12 @ \$114 = <b><u>\$1,368</u></b> (12 @ \$114 per unit) <sup>e</sup>                                             |
|         |                      | <b><u>\$4,622</u></b>       |                                                                                                                   |

<sup>a</sup> \$100 per unit = (\$2,500 inventory balance ÷ 25 units in inventory).

<sup>b</sup> \$100 per unit = (\$500 inventory balance ÷ 5 units in inventory).

<sup>c</sup> \$112 per unit = (\$2,800 inventory balance ÷ 25 units in inventory).

<sup>d</sup> \$114 per unit = (\$3,990 inventory balance ÷ 35 units in inventory).

<sup>e</sup> \$114 per unit = (\$1,368 inventory balance ÷ 12 units in inventory).

# Learning Objective

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A1:

Analyze the effects of  
inventory methods for both  
financial and tax reporting.

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# Financial Statement Effects of Inventory Costing Methods

Exhibit  
6.8

Because prices change, inventory methods nearly always assign different cost amounts.

| TREKKING COMPANY<br>For Month Ended August 31 |                            |          |          |                     |
|-----------------------------------------------|----------------------------|----------|----------|---------------------|
|                                               | Specific<br>Identification | FIFO     | LIFO     | Weighted<br>Average |
| <b><u>Income Statement:</u></b>               |                            |          |          |                     |
| Sales                                         | \$ 6,050                   | \$ 6,050 | \$ 6,050 | \$ 6,050            |
| Cost of goods sold                            | 4,582                      | 4,570    | 4,730    | 4,622               |
| Gross profit                                  | 1,468                      | 1,480    | 1,320    | 1,428               |
| Operating expenses                            | 450                        | 450      | 450      | 450                 |
| Income before taxes                           | \$ 1,018                   | \$ 1,030 | \$ 870   | \$ 978              |
| Income tax expense (30%)                      | 305                        | 309      | 261      | 293                 |
| Net income                                    | \$ 713                     | \$ 721   | \$ 609   | \$ 685              |
| <b><u>Balance Sheet:</u></b>                  |                            |          |          |                     |
| Inventory                                     | \$ 1,408                   | \$ 1,420 | \$ 1,260 | \$ 1,368            |

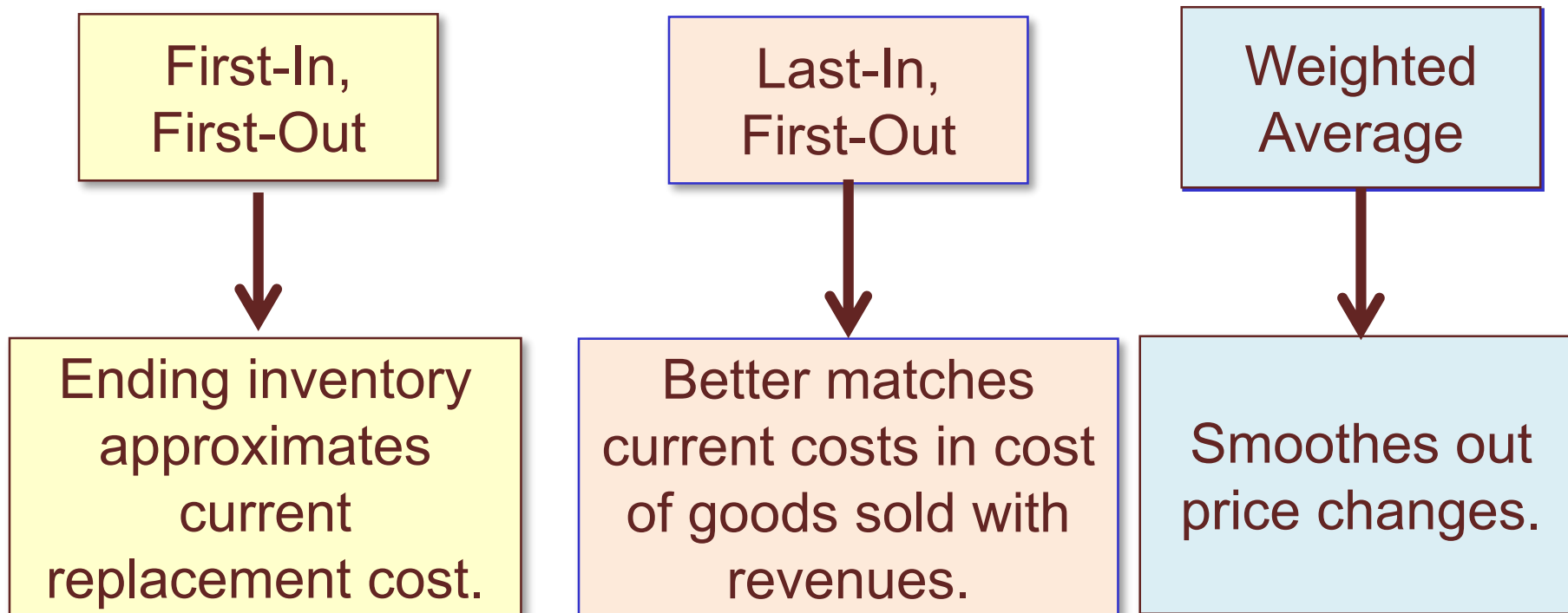
# Financial Statement Effects of Costing Methods

Because prices change, inventory methods nearly always assign different cost amounts.

| TREKKING COMPANY          |                |          |          |                     |
|---------------------------|----------------|----------|----------|---------------------|
| For Month Ended August 31 |                |          |          |                     |
|                           | Specific       |          |          |                     |
| <u>Income Statement:</u>  | Identification | FIFO     | LIFO     | Weighted<br>Average |
| Sales                     | \$ 6,050       | \$ 6,050 | \$ 6,050 | \$ 6,050            |
| Cost of goods sold        | 4,582          | 4,570    | 4,868    | 4,683               |
| Gross profit              | 1,468          | 1,480    | 1,182    | 1,367               |
| Operating expenses        | 450            | 450      | 450      | 450                 |
| Income before taxes       | \$ 1,018       | \$ 1,030 | \$ 732   | \$ 917              |
| Income tax expense (30%)  | 305            | 309      | 220      | 275                 |
| Net income                | \$ 713         | \$ 721   | \$ 512   | \$ 642              |
| <u>Balance Sheet:</u>     |                |          |          |                     |
| Inventory                 | \$ 1,408       | \$ 1,420 | \$ 1,122 | \$ 1,307            |

# Financial Statement Effects of Costing Methods

## Advantages of Methods



# Tax Effects of Costing Methods

The Internal Revenue Service (IRS) identifies several acceptable inventory costing methods for reporting taxable income.

If LIFO is used for tax purposes, the IRS requires it be used in financial statements.

# Consistency in Using Costing Methods

The **consistency concept** requires a company to use the same accounting methods period after period so that financial statements are comparable across periods.

# Learning Objective

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P2:

Compute the lower of cost or market amount of inventory.

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# Lower of Cost or Market

Inventory must be reported at market value when market is *lower* than cost.

Defined as current **replacement cost** (not sales price).  
Consistent with the conservatism principle.

Can be applied three ways:

- (1) separately to each individual item.
- (2) to major categories of assets.
- (3) to the whole inventory.

# Lower of Cost or Market

Exhibit  
6.9

A motor sports retailer has the following items in inventory:

| Inventory Items | Units | Per Unit |         | Total Cost       | Total Market | LCM Applied to Items    |
|-----------------|-------|----------|---------|------------------|--------------|-------------------------|
|                 |       | Cost     | Market  |                  |              |                         |
| → Roadster..... | 20    | \$8,500  | \$7,000 | \$170,000        | \$140,000    | \$ 140,000              |
| Sprint .....    | 10    | 5,000    | 6,000   | 50,000           | 60,000       | 50,000                  |
| → Totals .....  |       |          |         | <u>\$220,000</u> |              | <u><b>\$190,000</b></u> |

\$140,000 is the lower of \$170,000 or \$140,000.

The amount of \$190,000 is lower than the \$220,000 recorded cost.

# Learning Objective

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A2:

Analyze the effects of  
inventory errors on current  
and future financial  
statements.

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# Financial Statement Effects of Inventory Errors

## Income Statement Effects

Exhibit  
6.10

| Ending Inventory                                                                                    | Year 1                                                                                         |                                                                                                 | Year 2                                                                                          |                                                                                                 |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|                                                                                                     | Cost of Goods Sold                                                                             | Net Income                                                                                      | Cost of Goods Sold                                                                              | Net Income                                                                                      |
| Understated  ..... | Overstated   | Understated  | Understated  | Overstated   |
| Overstated*  ..... | Understated  | Overstated   | Overstated   | Understated  |

\* This error is less likely under a perpetual system versus a periodic because it implies more inventory than is recorded (or less shrinkage than expected). Management will normally follow up and discover and correct this error before it impacts any accounts.

# Financial Statement Effects of Inventory Errors

## Balance Sheet Effects

Exhibit  
6.12

| Ending Inventory                                                                                    | Assets                                                                                          | Equity                                                                                          |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Understated  ..... | Understated  | Understated  |
| Overstated  .....  | Overstated   | Overstated   |

# End of Chapter 6

# Learning Objective

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A3:

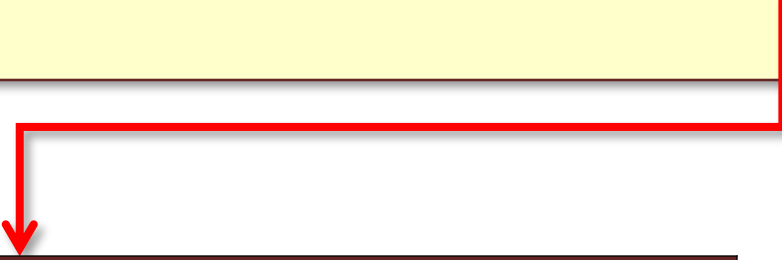
Assess inventory management using both inventory turnover and days' sales in inventory.

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# Inventory Turnover

Shows how many times a company turns over its inventory during a period. Indicator of how well management is controlling the amount of inventory available.

$$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Average inventory}}$$


$$\text{Average Inventory} = (\text{Beg. Inv.} + \text{End Inv.}) \div 2$$

# Days' Sales in Inventory

Reveals how much inventory is available in terms of the number of days' sales.

$$\text{Days' sales in inventory} = \frac{\text{Ending inventory}}{\text{Cost of goods sold}} \times 365$$

# Learning Objective

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P3 Appendix 6A:  
Compute inventory in a  
periodic system using the  
methods of specific  
identification, FIFO, LIFO, and  
weighted average.

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# Inventory Costing Illustration

## Periodic System

Exhibit  
6A.1

| Date    | Activity                  | Units Acquired at Cost            | Units Sold at Retail | Unit Inventory  |
|---------|---------------------------|-----------------------------------|----------------------|-----------------|
| Aug. 1  | Beginning Inventory. .... | 10 units @ \$ 91 = \$ 910         |                      | 10 units        |
| Aug. 3  | Purchases. ....           | 15 units @ \$106 = \$ 1,590       |                      | 25 units        |
| Aug. 14 | Sales. ....               |                                   | 20 units @ \$130     | 5 units         |
| Aug. 17 | Purchases. ....           | 20 units @ \$115 = \$ 2,300       |                      | 25 units        |
| Aug. 28 | Purchases. ....           | 10 units @ \$119 = \$ 1,190       |                      | 35 units        |
| Aug. 30 | Sales. ....               |                                   | 23 units @ \$150     | 12 units        |
|         | Totals .....              | <u>55 units</u><br><u>\$5,990</u> | <u>43 units</u>      | <u>12 units</u> |
|         |                           | Units available for sale          | Units sold           | Units left      |

# Inventory Costing Illustration

## Periodic System – Specification Identification



|                                                              |         |         |
|--------------------------------------------------------------|---------|---------|
| Total cost of 55 units available for sale                    |         | \$5,990 |
| Cost of goods sold:                                          |         |         |
| Aug 14 (8 @ \$91) + (12 @ \$106)                             | \$2,000 |         |
| Aug 30 (2 @ \$91) + (3 @ \$106) + (15 @ \$115) + (3 @ \$119) | 2,582   | 4,582   |
| Ending inventory                                             |         | \$1,408 |

On August 14, Trekking sold 8 bikes costing \$91 and 12 bikes costing \$106. On August 30, Trekking sold 23 bikes: 2 units costing \$91; 3 units costing \$106; 15 units costing \$115 and 3 units costing \$119.

# Inventory Costing Illustration

## Periodic System – Specification Identification

Here are the entries to record the purchases and sales.  
The numbers in red are determined by the cost flow assumption used.

All purchases and sales are made on credit. The selling price of inventory was as follows:

8/14    \$130  
8/31    \$150

|         |                       |       |       |
|---------|-----------------------|-------|-------|
| Aug. 3  | Purchases             | 1,590 |       |
|         | Accounts Payable      |       | 1,590 |
| Aug. 14 | Accounts Receivable   | 2,600 |       |
|         | Sales                 |       | 2,600 |
| Aug. 17 | Purchases             | 2,300 |       |
|         | Accounts Payable      |       | 2,300 |
| Aug. 28 | Purchases             | 1,190 |       |
|         | Accounts Payable      |       | 1,190 |
| Aug. 30 | Accounts Receivable   | 3,450 |       |
|         | Sales                 |       | 3,450 |
| Aug. 30 | Merchandise Inventory | 1,408 |       |
|         | Income Summary        |       | 498   |
|         | Merchandise Inventory |       | 910   |

# Inventory Costing Illustration

## Periodic System - FIFO



|                                           |          |          |
|-------------------------------------------|----------|----------|
| Total cost of 55 units available for sale |          | \$ 5,990 |
| Less ending inventory priced using FIFO:  |          |          |
| 10 units from Aug 28 purchase at \$119    | \$ 1,190 |          |
| 2 units from Aug 17 purchase at \$115     | 230      |          |
| Ending inventory                          |          | 1,420    |
| Cost of goods sold                        |          | \$4,570  |

During August, Trekking had a total of 55 units available for sale. They had 12 units in ending inventory and sold 43 units. The cost of the 12 bikes in ending inventory will be valued using the most recent purchases since the earlier units are assumed sold under FIFO.

# Inventory Costing Illustration

## Periodic System - LIFO



|                                              |        |          |
|----------------------------------------------|--------|----------|
| Total cost of 55 units available for sale    |        | \$ 5,990 |
| Less ending inventory priced using LIFO:     |        |          |
| 10 units from Aug 1 beginning inventory \$91 | \$ 910 |          |
| 2 units from Aug 3 purchase at \$106         | 212    |          |
| Ending inventory                             |        | 1,122    |
| Cost of goods sold                           |        | \$ 4,868 |

During August, Trekking had a total of 55 units available for sale. They had 12 units in ending inventory and sold 43 units. The cost of the 12 bikes in ending inventory will be valued using the earliest purchases since the most recently purchased units are assumed sold under LIFO.

# Inventory Costing Illustration

## Periodic System – Weighted Average



When a unit is sold, the **average cost** of each unit in inventory is assigned to cost of goods sold.



# Weighted Average – Determining Average Cost Per Unit



Step 1

|                 |                |
|-----------------|----------------|
| 10 units @ \$91 | \$ 910         |
| 15 units @ 106  | 1,590          |
| 20 units @ 115  | 2,300          |
| 10 units @ 119  | 1,190          |
| <b>55</b>       | <b>\$5,990</b> |

Step 2

|                    |                                                 |
|--------------------|-------------------------------------------------|
| \$5,990 / 55 units | <b>\$ 108.91</b><br>weighted avg. cost per unit |
|--------------------|-------------------------------------------------|

Step 3

|                                                     |                                                      |
|-----------------------------------------------------|------------------------------------------------------|
| Total cost of 55 units available for sale           | \$5,990                                              |
| Less <b>ending inventory</b><br>12 units @ \$108.91 | <b>\$1,307</b>                                       |
| <b>Cost of goods sold</b>                           | <b>\$4,683</b><br><b>(43 units at \$108.91 each)</b> |

# Learning Objective

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P4 Appendix 6B:  
Apply both the retail  
inventory and gross profit  
methods to estimate inventory.

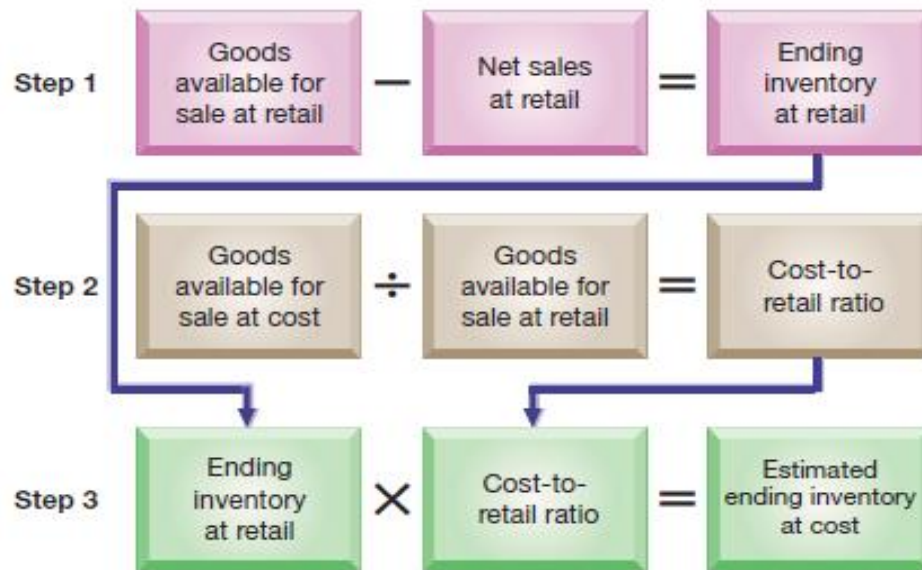
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# Appendix 6B:

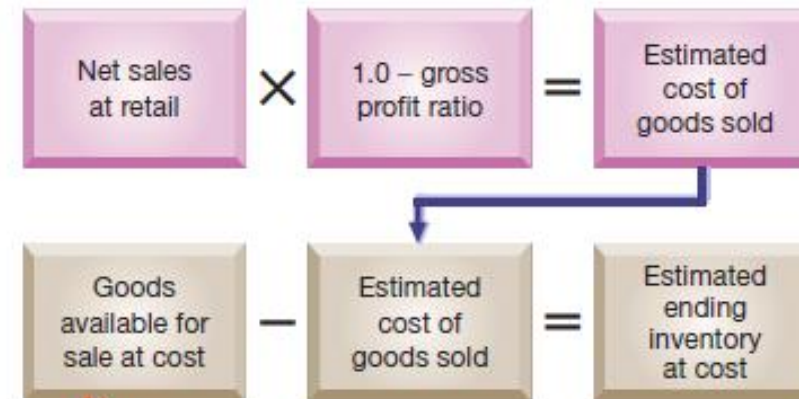
## Inventory Estimation Methods

Inventory sometimes requires estimation for interim statements or if some casualty such as fire or flood makes taking a physical count impossible.

### Retail Inventory Method



### Gross Profit Method



Exhibits  
6B.1 & 6B3

## NEED-TO-KNOW 6-3

A company has the following products in its ending inventory. (a) Compute the lower of cost or market for its inventory when applied separately to each product. (b) If the LCM amount is less than the recorded cost of the inventory, then record the December 31 LCM adjustment to the Merchandise Inventory account.

|                | Units | Per Unit |        | Total           |         | LCM<br>By item  |
|----------------|-------|----------|--------|-----------------|---------|-----------------|
|                |       | Cost     | Market | Cost            | Market  |                 |
| Road bikes     | 5     | \$1,000  | \$800  | \$5,000         | \$4,000 | \$4,000         |
| Mountain bikes | 4     | 500      | 600    | 2,000           | 2,400   | 2,000           |
| Town bikes     | 10    | 400      | 450    | 4,000           | 4,500   | 4,000           |
| <b>Total</b>   |       |          |        | <u>\$11,000</u> |         | <u>\$10,000</u> |

| Date    | General Journal       | Debit | Credit |
|---------|-----------------------|-------|--------|
| Dec. 31 | Cost of Goods Sold    | 1,000 |        |
|         | Merchandise Inventory |       | 1,000  |
|         |                       |       |        |